# Watcharapon Thodraksa

AI/ML Engineer · Computer Vision, Remote Sensing, MLOps

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## SUMMARY

AI/ML engineer who builds computer vision and satellite models for a Thai sugar mill and runs them in production. I handle the whole job: data and labels, training, evaluation on unseen data, the serving API, Kubernetes deployment and monitoring. Background in backend, DevOps and data from a full-stack co-op year.

## EXPERIENCE

**AI/ML Engineer (Contract), TRR** 2026 – Present

*Computer vision and satellite ML for a sugar mill group in Thailand*

- **CaneSight (CCTV dump-yard monitoring):** built the system that turns every truck dump into a result card for the mill's app: dust level 0–3, leafy-cane check, burnt or fresh, video clip and the matched truck. Seven top cameras and one side camera, served through a REST API, running on a CPU-only k3s server inside the mill (OpenVINO, ONNX).
- **MLOps for CaneSight:** MLflow model registry on k3s, training on Kaggle with pass/fail gates written before each test, promotion by registry alias with human approval, automatic deploy with checksum check and rollback, Flux GitOps with GitHub Actions checks, daily score-drift monitoring and Telegram alerts.
- **Burnt vs fresh cane classifier:** EfficientNet-B0 trained on labels from 25,980 weighbridge tickets; AUC 0.994 on 483 trucks from days the model never saw.
- **Leafy-cane detection:** fine-tuned DINOv2; correct verdict on 23 of 23 held-out truck frames at 1.8 s per image on CPU. Rejected a newer candidate that failed its gate on 300 new trucks.
- **CropScan (harvest monitoring from satellite):** Sentinel-2 with Sentinel-1 radar fallback under cloud; F1 0.883 on 8,924 held-out fields (FastAPI, rasterio, Railway).
- **Yield Pro (field health and yield):** SAR-to-NDVI gap-filling model with 95.7% of values within ±0.10 on 129k held-out pixels; yield ranges per field (LightGBM, PyTorch, MLflow).
- **CaneGate (weighbridge truck inspection):** cane grading and Thai licence-plate OCR (0 wrong reads on 60 hard images), shipped as Docker images and used at the mill.

**Software Developer (Co-op), Sycapt Co., Ltd.** Jan 2025 – May 2025

- Built an on-premises RAG knowledge system for a network with no internet access: LangGraph agents, BGE-M3 embeddings, pgvector hybrid search with re-ranking, Ollama; deployed on Kubernetes (Next.js, NestJS, Python).
- Ran four LLMs on a single 16 GB GPU by swapping models through 120 GB of RAM.
- Developed backend services for a payment service provider handling 1M+ transactions per day (PromptPay, Alipay).
- Built a map dashboard for 15,423 payment terminals nationwide (Leaflet, FastAPI) and an event-driven Kafka connector microservice (Java 21, Spring Boot, ELK).
- Designed the GitLab CI/CD pipeline: build, test, Docker build, staging, production with automatic rollback on Kubernetes.

## OTHER WORK

- **YouTube Shorts automation:** pipeline that reads channel analytics, picks topics, generates and uploads clips on schedule; 3.34M views across 189 clips.
- **Invited lecturer, Kasetsart University:** workshops on React Native with the Gemini API and on Git/GitHub.

## SKILLS

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| **ML / CV** | PyTorch, ONNX, OpenVINO, DINOv2, EfficientNet, YOLO, segmentation, OpenCV, LightGBM |
| **Remote sensing** | Sentinel-1 SAR, Sentinel-2 NDVI, rasterio, Planetary Computer |
| **MLOps / DevOps** | Kubernetes (k3s), Docker, MLflow, Flux GitOps, GitHub Actions, GitLab CI/CD, Prometheus, Grafana, Kaggle GPU |
| **LLM / RAG** | LangGraph, LangChain, pgvector, BGE-M3, Ollama |
| **Backend / Data** | Python, FastAPI, TypeScript, NestJS, Java Spring Boot, PostgreSQL, Kafka |

## EDUCATION

**B.Sc. Computer Science (Co-op Program), Kasetsart University, Chalermphrakiat Sakon Nakhon Campus** 2025

GPA 3.10

## AWARDS

1st Runner-up, KUSE AI Hackathon 2025 · Winner, UI Design-athon 2025 · Participant, RERU Cyber Hackathon 2025